

Year 8 Digital Technologies

Unit 3 — Cardbaord Arcade

St Brendan's College (Yeppoon), 2026



Unit overview		Cohort and/or class considerations
Start week: Term 3 Week 1 Finish week: Term 4 Week 9 In this unit, students explore physical computing by learning to program Arduino microcontrollers and apply these skills to the design and construction of interactive arcade machines. Students develop an understanding of algorithms, basic programming structures, and how hardware specifications influence performance and functionality. The design process is emphasised through the use of user stories and design criteria, guiding students to create solutions that meet specific user needs. Through iterative design, testing and refinement, students combine computational thinking with creativity, problem-solving and hands-on construction skills.		
Adjustments		
Student initials/identifier		
Year level/band against which the student is to be assessed	Band 7–8	
Category of need		
Adjustment/s and/or key information		
Review of adjustments		

Assessment 4	
Year 8	
Title	Hardware and Algorithms Exam
Description	In this assessment, students complete a written exam that assesses their understanding of foundational algorithms and hardware selection in the context of Arduino systems. Students demonstrate their ability to define and decompose problems, design and trace simple algorithms, and predict program outputs. The exam also evaluates students' understanding of how hardware specifications affect performance and how to select appropriate components for specific tasks. This assessment provides evidence of students' theoretical knowledge and computational thinking skills that underpin practical digital solutions.
Technique	Examination
Mode	Written
Conditions	• up to 50 minutes plus 10 minutes planning time • under supervised conditions • short response up to 75 words per item, up to 400 words for the task • extended response 100–200 words per item, up to 400 words for the task • 1–2 A3 pages or equivalent digital media that may include graphical representations with annotations
Timing	Week 6
Achievement standard	By the end of Year 8 students develop and modify creative digital solutions, decompose real-world problems, and evaluate alternative solutions against user stories and design criteria. Students acquire, interpret and model data with spreadsheets and represent data with integers and binary. They design and trace algorithms and implement them in a general-purpose programming language. Students select appropriate hardware for particular tasks, explain how data is transmitted and secured in networks, and identify cyber security threats. They select and use a range of digital tools efficiently and responsibly to create, locate and share content; and to plan, collaborate on and manage projects. Students manage their digital footprint.
Moderation	Cross marking to be completed within the department

Content descriptions	
Knowledge and understanding	Processes and production skills
Digital systems explain how hardware specifications affect performance and select appropriate hardware for particular tasks and workloads (AC9TDI8K01)	Investigating and defining define and decompose real-world problems with design criteria and by creating user stories (AC9TDI8P04)

Content descriptions	
Knowledge and understanding	Processes and production skills
	Generating and designing design algorithms involving nested control structures and represent them using flowcharts and pseudocode (AC9TDI8P05)
	trace algorithms to predict output for a given input and to identify errors (AC9TDI8P06)
	Producing and implementing implement, modify and debug programs involving control structures and functions in a general-purpose programming language (AC9TDI8P09)

Assessment 5	
Year 8	
Title	Arcade Machine
Description	In this assessment, students design and build a functioning cardboard arcade machine based on a provided user profile and set of user stories. Students apply the design process to define requirements, establish design criteria, and plan a solution that meets the needs of the target user. They design the user experience, develop and test algorithms, and implement programs using Arduino hardware and sensors. Students iteratively refine their designs, evaluate their solutions against the design criteria and user stories, and communicate their ideas using appropriate digital tools. The project demonstrates students' ability to integrate hardware selection, programming, creative design and project management to develop a complete digital solution.
Technique	Project
Mode	Multimodal
Conditions	2–4 A3 pages or equivalent digital media pages that may include annotated graphical representations
Timing	Week 9
Achievement standard	By the end of Year 8 students develop and modify creative digital solutions, decompose real-world problems, and evaluate alternative solutions against user stories and design criteria. Students acquire, interpret and model data with spreadsheets and represent data with integers and binary. They design and trace algorithms and implement them in a general-purpose programming language. Students select appropriate hardware for particular tasks, explain how data is transmitted and secured in networks, and identify cyber security threats. They select and use a range of digital tools efficiently and responsibly to create, locate and share content; and to plan, collaborate on and manage projects. Students manage their digital footprint.
Moderation	Monitoring: Week 6

Assessment 5

Conferencing: Week 8

Content descriptions	
Knowledge and understanding	Processes and production skills
Digital systems explain how hardware specifications affect performance and select appropriate hardware for particular tasks and workloads (AC9TDI8K01)	Investigating and defining define and decompose real-world problems with design criteria and by creating user stories (AC9TDI8P04)
investigate how data is transmitted and secured in wired and wireless networks including the internet (AC9TDI8K02)	Generating and designing design algorithms involving nested control structures and represent them using flowcharts and pseudocode (AC9TDI8P05)
	trace algorithms to predict output for a given input and to identify errors (AC9TDI8P06)
	design the user experience of a digital system (AC9TDI8P07)
	generate, modify, communicate and evaluate alternative designs (AC9TDI8P08)
	Producing and implementing implement, modify and debug programs involving control structures and functions in a general-purpose programming language (AC9TDI8P09)
	Evaluating evaluate existing and student solutions against the design criteria, user stories and possible future impact (AC9TDI8P10)
	Collaborating and managing select and use a range of digital tools efficiently, including unfamiliar features, to create, locate and communicate content, consistently applying common conventions (AC9TDI8P11)

Content descriptions	
Knowledge and understanding	Processes and production skills
	select and use a range of digital tools efficiently and responsibly to share content online, and plan and manage individual and collaborative agile projects (AC9TDI8P12)
General capabilities	
Critical and creative thinking	
- Inquiring - Develop questions - Identify, process and evaluate information - Generating - Create possibilities - Consider alternatives - Put ideas into action - Analysing - Interpret concepts and problems - Draw conclusions and provide reasons - Evaluate actions and outcomes - Reflecting - Transfer knowledge	
Digital literacy	
- Creating and exchanging - Plan - Create, communicate and collaborate - Respect intellectual property - Managing and operating - Manage content - Select and operate tools	

Teaching and learning sequence

Week	Planning details	Theory	Practical	Assessment
1	Key teaching and learning experiences	Students are introduced to digital systems and physical computing. They explore how hardware and software interact, with a focus on inputs, processes and outputs. Introduction to Arduino microcontrollers, including their purpose and real-world applications.	Students set up Arduino environments and build their first simple circuit (e.g. LED). They upload and run a basic program to control an output device.	Formative monitoring of student ability to correctly set up Arduino hardware and successfully upload a program. Teacher observation and questioning used to gauge understanding.
	Adjustments			
	Resources	Arduino IDE (or web editor) Introductory Arduino tutorials (official Arduino website) Teacher demonstration slides on digital systems		
2	Key teaching and learning experiences	Students learn about different types of inputs (buttons, sensors) and outputs (LEDs, buzzers). Introduction to basic programming concepts including variables and simple commands.	Students build circuits using buttons and LEDs. They write simple programs to control outputs based on inputs.	Formative monitoring through completion of set tasks. Students demonstrate ability to read input and produce an output.
	Adjustments			
	Resources	Arduino component kits Circuit diagrams and worksheets Online Arduino tutorials (inputs and outputs)		

Teaching and learning sequence				
3	Key teaching and learning experiences	Introduction to algorithms and control structures. Students learn about conditional logic (if/else statements) and how decisions are made within programs.	Students create programs that use conditionals to control outputs (e.g. LED turns on when button pressed). They begin linking logic to real-world interactions.	Teacher observation of student ability to implement conditional logic correctly. Debugging support provided as needed.
	Adjustments			
	Resources	Worked examples of pseudocode and simple algorithms Arduino tutorials on conditional statements Teacher-led demonstrations		
4	Key teaching and learning experiences	Students develop understanding of algorithms as step-by-step processes. They learn to represent algorithms using flowcharts and pseudocode.	Students design flowcharts for simple Arduino tasks, then implement them in code. They compare planned algorithms with actual program behaviour.	Collection and review of student-created flowcharts and corresponding programs.
	Adjustments			
	Resources	Flowchart templates Example algorithms and pseudocode Presentation slides on algorithm design		
5	Key teaching and learning experiences	Students learn about iteration (loops) and the importance of debugging. They explore how repetition is used in programs and strategies for identifying and fixing errors.	Students create programs using loops (e.g. flashing LED patterns). They test, debug and refine their code based on expected outputs.	Formative assessment through debugging tasks. Students demonstrate ability to identify and correct errors in code.
	Adjustments			
	Resources	Arduino tutorials on loops Debugging checklists Example faulty programs for troubleshooting		

Teaching and learning sequence				
6	Key teaching and learning experiences	Revision of key concepts including digital systems, hardware selection, algorithms, control structures and debugging. Students complete guided review activities and practice questions.	Students complete structured revision tasks, including tracing algorithms and predicting outputs.	Summative assessment: Hardware and Algorithms Exam conducted under supervised conditions.
	Adjustments			
	Resources	Revision worksheets Past/practice exam questions Teacher-prepared summary notes		
7	Key teaching and learning experiences	Students build on prior knowledge by exploring systems that use multiple inputs and outputs. They examine how more complex digital systems are structured and how components interact. Introduction to modular thinking in programming.	Students construct circuits that incorporate multiple components (e.g. button + LED + buzzer). They write programs that respond to multiple inputs and control different outputs.	Formative monitoring of students' ability to manage multiple inputs/outputs within a single program. Teacher observation and questioning.
	Adjustments			
	Resources	Arduino component kits (expanded components) Circuit diagrams for multi-component systems Arduino tutorials on multiple inputs/outputs		
8	Key teaching and learning experiences	Students are introduced to functions and modular programming. They learn how breaking programs into smaller parts improves readability, efficiency and debugging.	Students refactor previous programs to include functions. They develop structured programs that reuse code for repeated behaviours.	Teacher review of student programs to assess correct use of functions and code organisation.
	Adjustments			
	Resources	Worked examples of modular code Arduino tutorials on functions		

Teaching and learning sequence				
		Teacher modelling of code refactoring		
9	Key teaching and learning experiences	Students are introduced to the design process, focusing on user-centred design. They explore user profiles, user stories, and design criteria. Discussion of what makes an engaging arcade game.	Students analyse example arcade games and identify key features. They begin developing simple game ideas and draft initial user stories.	Collection and feedback on initial user stories and game concepts.
	Adjustments			
	Resources	Example user profiles and user stories Arcade game examples (videos or playable demos) Design process scaffolds		
10	Key teaching and learning experiences	Students focus on designing algorithms for their game ideas. They learn to map interactions and game logic using flowcharts and pseudocode, including nested control structures.	Students develop flowcharts and pseudocode for their proposed arcade games. They begin planning how inputs, outputs and timing will function within their system.	Teacher feedback on algorithm designs to ensure feasibility and alignment with user stories.
	Adjustments			
	Resources	Flowchart and pseudocode templates Worked examples of game logic Teacher modelling of algorithm design		
11	Key teaching and learning experiences	Students are introduced to Assessment 5: Arcade Machine project requirements, including success criteria and expectations. They revisit hardware selection and design constraints (including working with cardboard structures).	Students finalise their game concept, user stories and design criteria. They select components and begin planning both the physical layout and digital system.	Assessment 5 issued. Monitoring of student planning documentation (user stories, design criteria, initial sketches).
	Adjustments			
	Resources	Assessment task sheet and marking guide		

Teaching and learning sequence				
		Planning templates (design, user stories, criteria) Examples of simple arcade machine layouts		
12	Key teaching and learning experiences	Students explore prototyping as part of the design process, focusing on testing key interactions before full construction. Emphasis on iterative development and refinement.	Students build and test core components of their game using Arduino (e.g. scoring system, button inputs, feedback outputs). They begin implementing and refining their algorithms in code.	Teacher monitoring of prototype functionality and progress. Informal check-in against design plans.
	Adjustments			
	Resources	Arduino kits and components Prototyping guides and examples Debugging support materials		
13	Key teaching and learning experiences	Students examine principles of physical design, including stability, durability and usability. Discussion focuses on how physical structure impacts user experience and hardware integration, particularly when working with cardboard materials.	Students begin constructing the physical arcade machine using cardboard. They develop the structure, including housing for components, control interfaces and display areas.	Teacher monitoring of construction progress and alignment with design plans. Feedback provided on structural feasibility and usability.
	Adjustments			
	Resources	Cardboard construction materials (cardboard, cutters, tape, glue) Example arcade machine layouts and designs Safety guidelines for tool use		
14	Key teaching and learning experiences	Students explore how hardware components are embedded into physical systems. Focus on wiring management,	Students install Arduino components into their arcade machines. They connect inputs and outputs within the structure	Teacher observation of correct hardware integration and safe setup. Ongoing feedback on design implementation.

Teaching and learning sequence				
		component placement and maintaining accessibility for troubleshooting.	and begin aligning physical controls with programmed functions.	
	Adjustments			
	Resources	Arduino kits and wiring components Circuit diagrams and reference guides Teacher demonstrations of hardware integration		
15	Key teaching and learning experiences	Students revisit program structure and system integration, focusing on ensuring all components work together as a complete system. Emphasis on linking algorithms to real-world interaction.	Students continue developing and refining their code to match their arcade game design. They test full system functionality, ensuring inputs, outputs and game logic operate as intended.	Monitoring of progress against design criteria and functionality. Teacher feedback on code quality and system performance.
	Adjustments			
	Resources	Arduino programming guides Example integrated systems Debugging checklists		
16	Key teaching and learning experiences	Students explore systematic testing and debugging strategies. Focus on identifying faults in both hardware and software and applying structured approaches to resolve issues.	Students test their arcade machines, identify faults and refine both code and hardware. They document issues and solutions as part of the development process.	Teacher monitoring of debugging processes and problem-solving approaches.
	Adjustments			
	Resources	Debugging guides and checklists Example troubleshooting scenarios Teacher support materials		
17	Key teaching and learning experiences	Students examine the importance of user testing and feedback in the design process. They learn how to evaluate	Students conduct user testing with peers, gather feedback and refine their	Ongoing monitoring of improvements and alignment with user stories and design criteria.

Teaching and learning sequence				
		solutions against user stories and design criteria.	arcade machines to improve usability and functionality.	
	Adjustments			
	Resources	User testing feedback templates Design criteria checklists Peer review guidelines		
18	Key teaching and learning experiences	Students review requirements for submission and evaluation. Discussion focuses on effectively communicating design decisions and reflecting on the development process.	Students finalise their arcade machines and complete supporting documentation. They prepare their work for submission.	Summative assessment: Arcade Machine Project submission, including product and documentation.
	Adjustments			
	Resources	Assessment task sheet and marking guide Submission checklists Exemplar project documentation		
19	Key teaching and learning experiences	Students reflect on their learning across the unit, evaluating their solutions against design criteria, user stories and potential future improvements.	Students participate in a showcase of arcade machines, demonstrating their work and discussing their design and development process.	Final reflection and evaluation activities. Informal teacher assessment of student ability to articulate learning and outcomes.
	Adjustments			
	Resources	Reflection templates Evaluation frameworks Showcase materials and setup		

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